CS & IT ENGINEERING

Digital Logic

Boolean Theorems and Logical Gates

Lecture No. 01

By- Aditya sir

Topics to be Covered

Topic

1) Intro to Digital Logic
2) Cource Content
3) Intro to Boolean Theorems.

About Aditya Jain sir

"AJ Sir"

(Nagpur)

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)

9.09/10

4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science

TA/RA

6. Published multiple research papers in well known conferences along with the team

10/10

7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Telegram Channel

@ADITYASIR_PW

Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

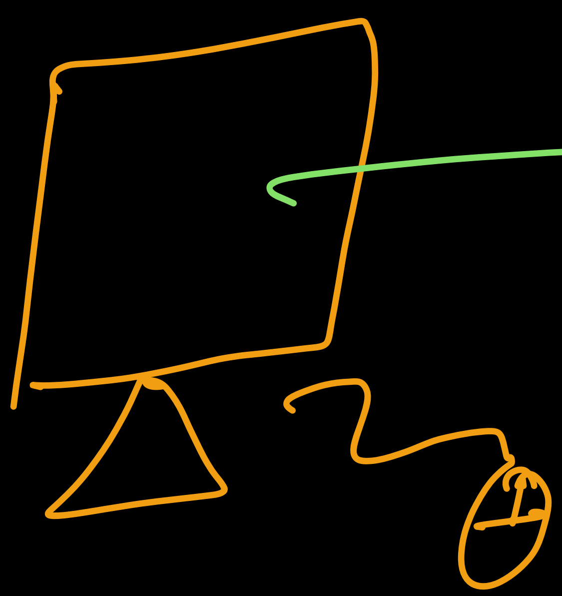
Course Content:

- Chap 1 :- Boolean Theorems & Logical Gates → 08
- Chap 2 :- Combinational Circuits → 11
- Chap 3 :- Sequential Circuits → 11
- Chap 4 :- Miscellenous Concepts (Number System) → 08

Boolean Theorems

True / False

1 / 0



0/1

Binary

Binary Number System

2 : [0 , 1]

Base : 2

Decimal System

10 : [0 – 9]

Base : 10

Digits/Number

Combination of digits → Number

[1 or more digit]

$(10)_{10}$

[0 – 9]

$\to$ base : 10 $\to$ decimal

Number : one number (ten)

Digits : 2 digits

$(1)_{10}$

$(9)_{10}$

number

Digits & Num

eg:-

$$(24)_{10}$$

more weightage $\quad 2 \quad 4 \quad$ least weightage.

$$\Downarrow$$

$$(20 + 4)$$

$$2 \times 10 + 4 \times 1$$

$$\left[\, 2 \times \underline{\underline{10^{1}}} + 4 \times \underline{\underline{10^{0}}} \,\right]$$

weights

$$(3)_{10} \longrightarrow 3\times10^{0} = 3\times1 = 3 \checkmark$$

$$(27)_{10} \longrightarrow 2\times10^{1} + 7\times10^{0} = 20+7 = \underline{\underline{27}}$$

weightage

base

$$(934)_{10} \longrightarrow 9\times10^{2} + 3\times10^{1} + 4\times10^{0}$$

$$900 + 30 + 4 = \underline{\underline{934}}$$

Binary

$(21)_2 \rightarrow$ binary

invalid

$[0 - 1]$

$(110)_{10} \rightarrow$

Binary → Decimal

$[2^2 \quad 2^1 \quad 2^0]$ wt

eg1 :-

$(101)_2 \rightarrow$ 1 0 1

$1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$= 1 \times 4 + 0 \times 2 + 1 \times 1$

$= 4 + 0 + 1 = (5)_{10}$

$(101)_2 \rightarrow (5)_{10}$

Classwork :-

1 0

Shortcut

1 → cut

0 → cut ✗

1) $(1001)_2 \Rightarrow 1\times2^3 + 0\times2^2 + 0\times2^1 + 1\times2^0 = 8+0+0+1$
$= 9$

2) $(010010)_2 \Rightarrow 2^4 + 2^1 = 16+2 = 18$

3) $(11111)_2 \rightarrow 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 1+2+4+8+16$
$= 31$

4) $(10101)_2 \rightarrow 2^4 + 2^2 + 2^0$
$= 16+4+1 = 21$

$$P \to 0,\ 1 \qquad Q \to 0,\ 1$$

2^1 P	2^0 Q		Decimal
0	0	$\rightarrow$	0
0	1	$\rightarrow$	1
1	0	$\rightarrow$	2
1	1	$\rightarrow$	3

2^n possiblitis

$$\begin{array}{ccccc} 2^2 & 2^1 & 2^0 & & \\ P & Q & R & & \\ 0 & 0 & 0 & \rightarrow & 0 \\ 0 & 0 & 1 & \rightarrow & 1 \\ 0 & 1 & 0 & \rightarrow & 2 \\ 0 & 1 & 1 & \rightarrow & 3 \\ 1 & 0 & 0 & \rightarrow & 4 \\ 1 & 0 & 1 & \rightarrow & 5 \\ 1 & 1 & 0 & \rightarrow & 6 \\ 1 & 1 & 1 & \rightarrow & 7 \end{array}$$

$$\underbrace{3 \text{ digits} \rightarrow 2 \text{ possiblity}}$$

$$2 \times 2 \times 2 = 2^3 = 8$$

Max number with n digits $(\text{Base}^n - 1)$

Decimal		Binary		Decimal
0 – 9	$\rightarrow (10^1 - 1)$	$(1)_2$	$\rightarrow (2^1 - 1) =$	1
0 – 99	$\rightarrow (10^2 - 1)$	$(11)_2$	$\rightarrow (2^2 - 1) \rightarrow$	3
0 – 999	$\rightarrow (10^3 - 1)$	$(111)_2$	$\rightarrow (2^3 - 1) \rightarrow$	7
0 – 9999	$\rightarrow (10^4 - 1)$	$(1111)_2$	$\rightarrow (2^4 - 1) \rightarrow$	15

every digit is max digit

$\begin{bmatrix} \overset{2^3}{P} & \overset{2^2}{Q} & \overset{2^1}{R} & \overset{2^0}{S} \end{bmatrix} \longrightarrow 2^4 = \underline{16 \text{ Combi}}$

P	Q	R	S	
0	0	0	0	→ 0
0	0	0	1	→ 1
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	→ 5

n variables (binary digits)

→ 2^n Combinations

→ min value → 0

→ max value → $(2^n - 1)$

Boolean Theorems :

→ ① Logical 'OR' operation (+)

Binary variable

$(x + 0 = x)$

1) $A + 0 = A$

2) $A + 1 = 1$

3) $1 + A = 1$

4) $A + A = A$

(anything + 1 = 1)

$0 + 0 = 0$

$1 + 1 = 1$

A

0 1

$\bar{A} \longrightarrow$ (Complement of A)

A		$\bar{A}$
0	$\longleftrightarrow$	1
1	$\longleftrightarrow$	0

5) $\bar{A} + 0 = \bar{A}$

$x + 0 = x$

Let $\bar{A} = x$

$x + 0 = x$

$\rightarrow \quad A + A = A$

$B + B = B$

$\bar{A} + \bar{A} = \bar{A}$

$\rightarrow A + A = A$

$\rightarrow A + A + A = A$

$A + A$

$A + A + A + A = A$

* $$A + \bar{A} = 1$$

$$1 + \text{anything} = 1$$

A

$0 \rightarrow \bar{A} = 1$

$1 \rightarrow \bar{A} = 0$

→ $$\bar{A} + A = 1$$

$$\boxed{0 + 0 = 0}$$

$$\boxed{\begin{aligned} 1 + 0 &= 1 \\ 0 + 1 &= 1 \\ 1 + 1 &= 1 \end{aligned}}$$

$$AB + BCD + 1 + A = 1$$

$$1 + \overline{AB} + \overline{B}C + D = \underline{1}$$

$$ABC + ABC = ABC$$

$$\overline{AB} + \overline{AB} = \overline{AB}$$

$$PQ + \overline{PQ} = 1$$

$$\left.\begin{aligned} PQ &= x \\ \overline{PQ} &= \bar{x} \end{aligned}\right\} \longrightarrow \underline{\underline{x + \bar{x} = 1}}$$

$$\underline{\underline{\bar{A}B + C + 0}} = \underline{\underline{\bar{A}B + C}}$$

2) Logical 'AND' operator (.)

1) $A \cdot 0 = 0 = 0 \cdot A$

2) $0 \cdot \text{anything} = 0$ AND

3) $(AB + \bar{D}) \cdot 0 = 0$

4) $A \cdot 1 = A$

5) $\bar{A} \cdot 1 = \bar{A}$

$x \cdot 1 = x$

6) $A \cdot A = A$

7) $A \cdot A \cdot A =$

$A \cdot A = A$

$\bar{A} + A = 1$

1 0

8) $\bar{A} \cdot A = 0$

1 0

$\underline{\underline{AB}}$

$$\boxed{\begin{array}{c} AB \cdot \overline{AB} = 0 \\ x \cdot \bar{x} = 0 \end{array}}$$

2 mins Summary

Topic

Topic

Topic

Topic

Topic

Intro

Course Plan

Boolean Algebra & Number System

Intro

"9:30" PM

THANK - YOU